

# Painter Naming in Generated Art: Distributional Proximity, Scene Distinguishability, and Color Responsiveness

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## Abstract

Painter instructions can move generated images closer to a painter’s works while changing the variation within a generated collection. We investigate these changes using 31 interpretable image features and 70 digital reproductions by Monet and Cézanne. A first study comprises 1,006 images from three generation services, with shared scene descriptions and repeated, randomized prompt comparisons. Naming lowers primary energy discrepancy in all six service–painter comparisons; four reject the conditional randomization null after adjustment. Aggregate variation decreases consistently, but this does not uniformly imply less within-scene variation or poorer scene distinguishability. In a held-repetition diagnostic, FLUX/Monet scene retrieval improves from 44.4% to 59.7% despite substantial between-scene contraction. A second, 192-image experiment crosses artist-free, generic-painting and two painter-name instructions with muted and vivid palettes. The generic clause reduces the vivid-minus-muted median-chroma response by 0.853 development-IQR units relative to the artist-free control in a secondary comparison. Additional Monet and Cézanne effects are  $-0.284$  and  $-0.018$ , with simultaneous 95% family intervals  $[-0.585, 0.016]$  and  $[-0.343, 0.308]$ ; neither primary comparison is resolved. The findings distinguish distributional proximity, concentration, scene distinguishability and color response. Contraction alone does not establish poorer scene retrieval or attenuation beyond generic style language. They concern the delivered services and selected digital reference panels, rather than perceptual fidelity or an identified internal model mechanism.

## 1 Introduction

What does it mean for a generated collection to resemble a painter’s work? Individual images can evoke a familiar palette or motif while covering only a small part of the variation present across the painter’s paintings. Conversely, a collection can be diverse yet systematically differ from those paintings. Evaluating resemblance therefore requires more than distance to an average image or average feature vector. It requires comparing distributions and asking what kind of variation those distributions contain.

Painter naming offers a direct intervention with which to study this problem. Given the same description of a scene, adding “in the style of Claude Monet” changes the artistic instruction while leaving the written scene description intact. The resulting images need not change only in style: the instruction may also affect composition, depicted content or service behavior. Nevertheless, the comparison identifies how the complete generation system responds to an explicit painter reference under a common prompt design.

We examine three related questions. Does painter naming move generated feature distributions closer to paintings by that painter? If the generated distribution contracts, do different requested scenes become less distinguishable? Finally, does naming reduce the response to a controlled visual instruction beyond the effect of generic painting language? The second and third questions test a possible interpretation of contraction as reduced responsiveness; that interpretation does not follow from a smaller covariance trace alone.

Two studies address these questions for Monet and Cézanne. Study 1 compares 1,006 images from three services using 24 shared scene descriptions and three repetitions per condition. We measure distributional discrepancy and decompose variation within and between descriptions, then test scene retrieval with held-out repetitions. Study 2 uses 192 new images from one service to cross four style instructions with muted and vivid palettes. Comparing a painter clause with a generic painting-style clause tests whether the named clause changes the response beyond this generic clause. Both studies use the same fixed interpretable representation and a 70-work digital reference panel.

The resulting account separates proximity, variation and responsiveness. Closer distributions can be narrower without losing scene distinguishability, and response decreases relative to artist-free prompts also occur under a generic style clause. These distinctions limit what contraction alone establishes about the original/generated gap. Our target is the behavior of the delivered image services in selected painter cases, not the frequency of a phenomenon across art history or a complete measure of style.

## 2 Related work

Quantitative studies of art have used color statistics, image structure and information-theoretic measures to characterize variation across painting collections (Kim et al., 2014; Sigaki et al., 2018; Lee et al., 2020). More recently, Kim et al. (2026) contrasted autoencoder and CLIP representations and tested context-keyword image-to-image transformations in a study of art-historical development. We instead use text-only generation to examine how a painter-name intervention changes the distribution of interpretable features relative to a finite set of that painter’s works.

Research on generated art has also examined artistic imitation and attribution. AI-Pastiche assesses generated paintings through human authenticity and prompt-adherence surveys (Asperti et al., 2025). AI-WikiArt uses captions of original paintings to condition generation and evaluates attribution with vision-language models (Fu et al., 2025). These approaches address judgments about individual images and their attribution. Our distribution-level analysis complements them by distinguishing feature proximity from variation across and within scene descriptions; it does not replace human evaluation.

The preprint by Asperti (2026) reports separation between human and generated images in CLIP space. Generative-model evaluation also distinguishes fidelity from diversity and coverage (Naeem et al., 2020), while preprocessing can substantially affect feature-based comparisons (Parmar et al., 2022). We apply this distinction to painter naming and use energy discrepancy (Székely and Rizzo, 2013) to compare full empirical distributions without reducing either collection to its centroid.

Compositional generation benchmarks separately evaluate attribute binding and object relations, including color binding (Huang et al., 2023). Our palette intervention measures the global chroma response to an explicit instruction. It does not score whether a requested color is assigned to a particular object. Together with held-repetition scene retrieval, it tests observable implications of reduced responsiveness without equating them with semantic adherence or an internal conditioning mechanism.

## 3 Methods

### 3.1 Study 1: reference paintings and generation design

The reference panel contains 38 Monet and 32 Cézanne works, represented by Wikimedia-delivered digital reproductions. The painters were selected after exploratory work on four

painters; those exploratory results are not pooled into this analysis. Reference identities were screened against prior project exposure records. The available panel is not a probability sample of either painter’s oeuvre. Selection depended on image availability and landscape eligibility, with one figure-led work excluded during visual coding.

Three broad content classes organize the comparison: water-dominated scenes, built scenes and land-dominated scenes. Their reference counts are 21/4/13 for Monet and 3/11/18 for Cézanne. Coding was performed through a maintainer-run LLM visual assessment, without independent human adjudication; 17 mixed or uncertain annotations remain flagged. These classes provide coarse content adjustment. They do not match each generated image to a specific original composition.

The prompt inventory contains eight descriptions per content class. Each is used for both painters, with three repetitions, giving 72 intended images per service–painter–condition cell. The artist-free prompt begins “Create an oil painting on canvas.” The named prompt adds “In the style of [painter].” Both then give the identical detailed scene description and request only the painting area, without a frame, signature, letters or watermark. No original painting is provided to the generator. A third, short-scene named condition on one service replaces the detailed description with a short scene description; this changes content specificity as well as prompt length. An exact example appears in Appendix A.

The commercial routes request Google’s `gemini-3.1-flash-image`, termed Nano Banana 2 (NB2), and Black Forest Labs’ `flux.2-max` (FLUX), through fixed OpenRouter providers. A local OAuth image service requests `gpt-image-2`, termed the OAuth service below. These are requested service identifiers, not attestations of immutable model weights. NB2 and FLUX return 1024-by-1024 images, in JPEG and PNG respectively. The OAuth service returns PNGs with variable geometry and reported quality: 424 of 430 images report low quality and six report medium, all six in artist-free conditions. Its prompt comparison therefore includes possible changes in delivered quality.

Condition order is randomized independently within each service, painter, description and repetition group. For the OAuth service, all three conditions are randomly ordered. Repetitions are distributed across eight assigned collection batches; the realized collection spans approximately 11.5 hours. Batches do not represent independent backend sessions. The 1,008 assigned slots yield 1,006 images: 288 each for NB2 and FLUX, and 430 for OAuth. Two Cézanne/OAuth/short-scene requests are refused and remain missing. One transient FLUX failure is recovered by an exact-payload technical retry. Successful images are retained without selecting among alternatives or screening for stylistic resemblance. Generated content classes refer to the intended prompts; adherence has not been independently assessed.

### 3.2 Feature representation and distribution measures

Each image is measured in 31 interpretable coordinates: 11 color, eight spatial and 12 digital-texture features (Appendix A). The primary pipeline applies orientation metadata, converts valid embedded color profiles to sRGB, assumes sRGB for compatible unprofiled files and preserves aspect ratio at a 512-pixel short side. Images are opaque and resized with Lanczos filtering, without upsampling or discretionary cropping. Each coordinate is standardized by its median and interquartile range (IQR) in a separate development collection of 221 works from four painters, with equal total weight per development painter. Neither current references nor generated images fit this transformation.

For reference vectors  $X = \{x_i\}$  and generated vectors  $Y = \{y_j\}$ , let  $a_i, b_j$  be nonnegative weights summing to one within each set. We use the empirical V-statistic energy discrepancy

$$\begin{aligned}\widehat{\mathcal{E}}(X, Y) = & 2 \sum_{i,j} a_i b_j \|x_i - y_j\|_2 - \sum_{i,k} a_i a_k \|x_i - x_k\|_2 \\ & - \sum_{j,l} b_j b_l \|y_j - y_l\|_2.\end{aligned}\tag{1}$$

This statistic incorporates distances within and between the two collections. Its finite-sample baseline need not be zero for independent samples from the same population. We interpret lower values as closer empirical distributions in the specified feature geometry, not as a calibrated degree of artistic similarity.

References have equal weight,  $a_i = 1/N_p$ . If painter  $p$  has  $N_{pc}$  reference works in class  $c$  and a generated cell has  $n_{pc}$  available images in that class, a generated image receives  $b_j = N_{pc}/(N_p n_{pc})$ . Thus both distributions have the reference panel’s class proportions. This weighting reduces differences in broad content mixture but leaves within-class content differences unresolved.

To measure spread, we use the weighted population-form covariance trace

$$\mathcal{V}(Y) = \sum_j b_j \|y_j - \bar{y}\|_2^2, \quad \bar{y} = \sum_j b_j y_j.\tag{2}$$

We distinguish the generated/reference ratio  $\mathcal{V}(Y)/\mathcal{V}(X)$  from the named/artist-free ratio  $\mathcal{V}(Y_N)/\mathcal{V}(Y_F)$ . A value below one has a different comparison target in these two ratios. Coordinate-wise squared IQR sums provide a complementary spread summary.

Repeated descriptions allow an exact descriptive decomposition. For description  $b$ , let  $w_b$  be its total image weight and  $\bar{y}_b$  its weighted mean. Then

$$\mathcal{V}(Y) = \underbrace{\sum_b \sum_{j \in b} b_j \|y_j - \bar{y}_b\|_2^2}_W + \underbrace{\sum_b w_b \|\bar{y}_b - \bar{y}\|_2^2}_B.\tag{3}$$

$W$ : within descriptions       $B$ : between description means

The between-description term can be further separated into within-class and between-class components. With three repetitions per description, these are finite-set summaries, not bias-corrected estimates of population variance components. In particular, the between-description term also includes noise in means estimated from only three images.

### 3.3 Study 1: prompt contrasts and inference

The prespecified inferential family comprises six named-minus-artist-free energy contrasts and two detailed-minus-short-scene named contrasts on OAuth. A contrast uses complete pairs sharing the description, repetition and assigned batch, with the same content weights in both conditions. The paired statistic retains all terms in Equation 1; it is not a difference of average distances to a reference mean. Appendix B gives the swap identity.

For each contrast, 99,999 Monte Carlo sign assignments yield a two-sided randomization  $p$ -value with the plus-one correction. Holm adjustment controls the fixed family of eight tests at level .05. The sharp null states that prompt assignment changes neither availability nor measured features in any assigned position, under the fixed request and technical-retry policy and no interference. For a pairwise OAuth comparison, the third condition’s position is conditioned on. The tests concern this finite design and reference panel. They do not test an oeuvre-wide mean effect, perceptual equivalence or equality of original and generated population distributions. Treatment-dependent call duration or service carryover could violate no interference.

The Study 1 prompt inventory, primary metrics and eight-test family were specified before its evaluation features were measured. The subsequent feature-view, variance-decomposition,

reference-influence, painter-interaction, retrieval and detection/coverage diagnostics are post-result analyses. They are descriptive and introduce no additional confirmatory tests. Study 2 has a separately specified two-contrast family, described below.

### 3.4 Measurement and reference diagnostics

We examine three processing pipelines: the primary 512-pixel pipeline, a 256-pixel version and common JPEG encoding at quality 90 with 4:2:0 subsampling at 512 pixels. Each uses its own development-only scaler. Four feature views cross these pipelines: all 31 coordinates; 28 coordinates after dropping three derived multiscale summaries; 31 coordinates divided by the square root of their family size; and the 19 color/spatial coordinates with texture removed. The family-rescaled squared distance sums mean squared differences within each family; it does not equalize observed family variance or perceptual importance. This gives 12 views per service–painter comparison.

Reference diagnostics delete one work, holding-collection membership group or content class at a time. Collection groups use exact recorded sets of holding- collection IDs and do not identify photographic capture workflows. Work and collection-group deletions preserve the original content-class masses while reweighting the remaining works within each class; class deletion renormalizes the surviving class masses. A separate analysis refits the scaler after omitting each development painter.

### 3.5 Scene distinguishability under contraction

Using the Study 1 detailed named and artist-free conditions, we classify each held-out repetition by its nearest scene centroid in the fixed feature space. For each service, painter and condition, centroids use the other two repetitions of each scene; the held-out image never enters its own or any competing centroid. Cycling through the three repetitions yields 72 queries per condition. We evaluate both all 24 scenes and the eight scenes in the query’s broad content class. The latter comparison removes discrimination attributable solely to broad class. The scaler remains fixed on development paintings.

Retrieval uses Euclidean distances in all 31 coordinates, color alone, spatial features alone, texture alone and the 19 non-texture coordinates, under each processing pipeline. Every scene has equal weight. For this diagnostic, variance components are also recomputed with equal scene weights; these differ from the reference-class-weighted components in Equation 3. The comparisons are descriptive. Uniform positive rescaling of a condition leaves nearest-centroid assignments unchanged; retrieval therefore measures relative scene distinguishability, not absolute response gain. It also does not establish whether requested objects are present.

### 3.6 Study 2: controlled color responsiveness

Study 2 crosses six fixed landscape descriptions (two per broad class), four style arms, two palette instructions and four repetitions, yielding 192 images. All arms begin with the same oil-painting request. The artist-free arm adds no style clause; the generic arm adds “In a traditional landscape-painting style.”; the named arms add the Monet or Cézanne clause. The generic clause is a control for painting-style language, distinct from Study 1’s short-scene named condition. It is a single alternative clause, not a factorial isolation of artist identity or a representative sample of generic phrasings. The palette instruction requests either “a muted, low-chroma palette throughout the painting” or “a vivid, high-chroma palette throughout the painting.” Scene text and the closing instruction are otherwise identical. The free and generic outputs are shared controls for both painters, giving 48 images per arm.

The 24 scene/repetition blocks were randomly ordered, with all eight arm/palette requests jointly randomized within each block before generation. The same OAuth service identifier was used, requesting one opaque 1024-by-1024 PNG at medium quality. All 192 images were retained and measured under all three processing pipelines. Actual delivery differed from the requested settings: all outputs were nonsquare RGB PNGs, and 174 reported low quality and 18 medium despite a medium-quality request. The medium-quality counts were 10, 4, 3 and 1 in the free, generic, Monet and Cézanne arms. These post-request fields were neither filtered nor adjusted away; the estimand concerns the complete delivered-service response. Acquisition accounting for a separate incomplete collection is given in Appendix C.

The prespecified endpoint is median pixel CIELAB chroma in the primary 512-pixel pipeline. Its development-standardized value is

$$C_{\text{img}} = \text{median}_{\text{pixels}} \sqrt{(a^*)^2 + (b^*)^2}, \quad z = \frac{C_{\text{img}} - 11.24253}{7.34764}.$$

The center and IQR are fixed on the prior development collection. All cross-processing chroma comparisons use this same primary scale, rather than refitting or changing units. Let  $d(a)$  be the equal-scene mean vivid-minus-muted response for arm  $a$ . The two primary interactions are

$$\kappa_M = d(M) - d(G), \quad \kappa_C = d(C) - d(G), \quad (4)$$

where  $G$  denotes the generic arm and  $M, C$  the named arms. Four repeat-block contrasts per scene estimate each interaction; the two estimates share the same generic draws. Template-stratified covariance and Welch–Satterthwaite  $t$  approximations provide Bonferroni simultaneous 95% family intervals and Holm-adjusted two-sided tests for the two primary interactions. These are model-based inferences conditional on six fixed scenes, independent repeat-block errors and stable within-scene response distributions. Randomized dispatch does not make a weak zero-interaction test exact. Full formulas appear in Appendix C.

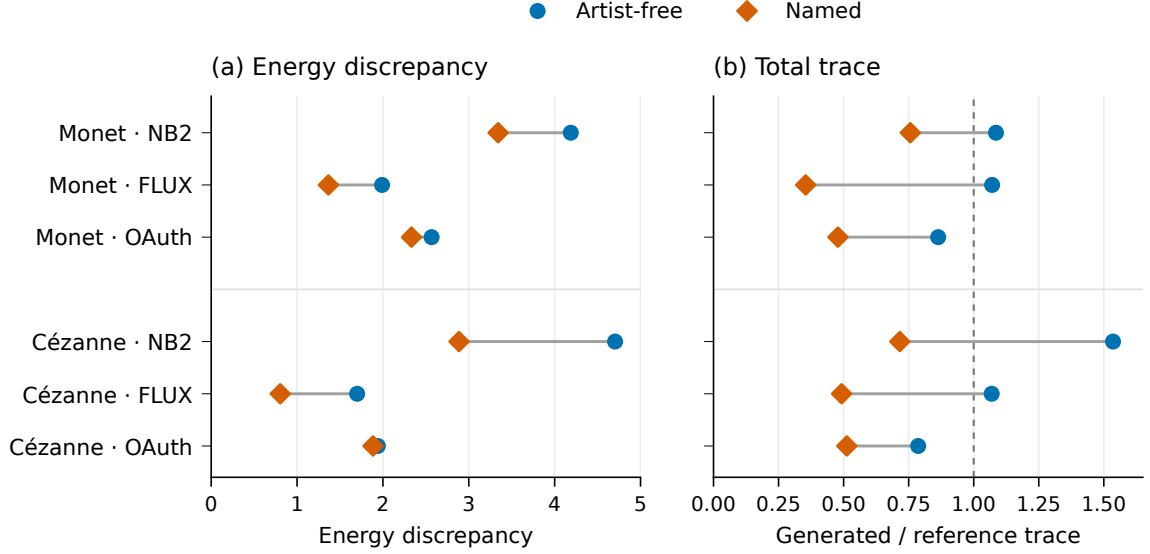
Control responses, named-minus-free and generic-minus-free contrasts use nominal 95% intervals as secondary summaries. A negative interaction supports a smaller positive chroma response only with positive control responses and no named response reversal. An interval including zero remains unresolved. Processing variants and the remaining coordinates are descriptive. The two primary interactions were specified before Study 2 generation and remain separate from Study 1’s eight-test family.

To connect responsiveness with original paintings, we compare each generated chroma distribution with the already exposed reference panel using one-dimensional Wasserstein distance, empirical quantiles and both directions of central-range occupancy. Quantiles select the first value whose exact rational cumulative weight reaches the specified probability. We report empirical and equal-class reference weights; generated distributions have equal class mass and, for the combined muted/vivid condition, equal polarity mass. The latter is a designed mixture, not a natural output distribution. These one-coordinate comparisons are descriptive and do not recalibrate the reference panel as a validated style target.

## 4 Results

### 4.1 Painter naming improves primary feature proximity

Painter naming lowers primary energy discrepancy in all six service–painter comparisons (Figure 1). NB2 decreases from 4.188 to 3.342 for Monet and from 4.704 to 2.886 for Cézanne; FLUX decreases from 1.990 to 1.365 and from 1.700 to 0.804, respectively. These four contrasts reject the conditional sharp null after adjustment across the eight-test family (Table 1). The OAuth differences are smaller, and neither survives adjustment. The detailed-versus-short-scene



**Figure 1:** Primary feature discrepancy and spread under artist-free and named prompting. Energy compares each generated distribution with its painter’s reference panel. Trace ratios compare generated spread with reference spread; the line at one denotes equal empirical trace. Each displayed generated cell contains 72 images; reference panels contain 38 Monet and 32 Cézanne works. The panels are descriptive and do not share a numerical scale.

| Comparison             | Service | Painter | Pairs | $\Delta\hat{\mathcal{E}}$ | Holm $p$ |
|------------------------|---------|---------|-------|---------------------------|----------|
| Named – free           | NB2     | Monet   | 72    | −0.846                    | .00310   |
|                        | NB2     | Cézanne | 72    | −1.818                    | .00008   |
|                        | FLUX    | Monet   | 72    | −0.625                    | .00008   |
|                        | FLUX    | Cézanne | 72    | −0.896                    | .00008   |
|                        | OAuth   | Monet   | 72    | −0.234                    | .19308   |
|                        | OAuth   | Cézanne | 72    | −0.056                    | 1.00000  |
| Detailed – short scene | OAuth   | Monet   | 72    | −0.037                    | 1.00000  |
|                        | OAuth   | Cézanne | 70    | −0.272                    | .09280   |

**Table 1:** The complete prespecified test family, using the primary 31-feature representation and matched pairs. Negative changes indicate smaller energy in the first condition. Values are conditional randomization results, with 99,999 sign draws and Holm correction across all eight comparisons. Both conditions in the final two rows name the painter.

comparisons likewise provide no rejection after adjustment. Non-rejection does not establish equivalence.

The same primary feature space also shows that the named collections have less aggregate spread than the references. Generated/reference trace ratios for the six detailed named cells range from .354 to .756, whereas the four NB2 and FLUX artist-free cells range from 1.069 to 1.535. Thus, reduced spread is not a general property of all generated images in this experiment. It is associated with the tested painter-naming conditions. The two short-scene named OAuth cells have still lower reference-relative ratios, .306 for Monet and .395 for Cézanne, although their energy contrasts do not establish an advantage of detailed prompting.

Energy and spread are distinct summaries but are not statistically independent: Equation 1 includes a within-generated distance term. These results therefore describe the joint behavior of two measurements, not evidence for a necessary trade-off between fidelity and diversity.



## 4.2 Aggregate contraction has different within-prompt components

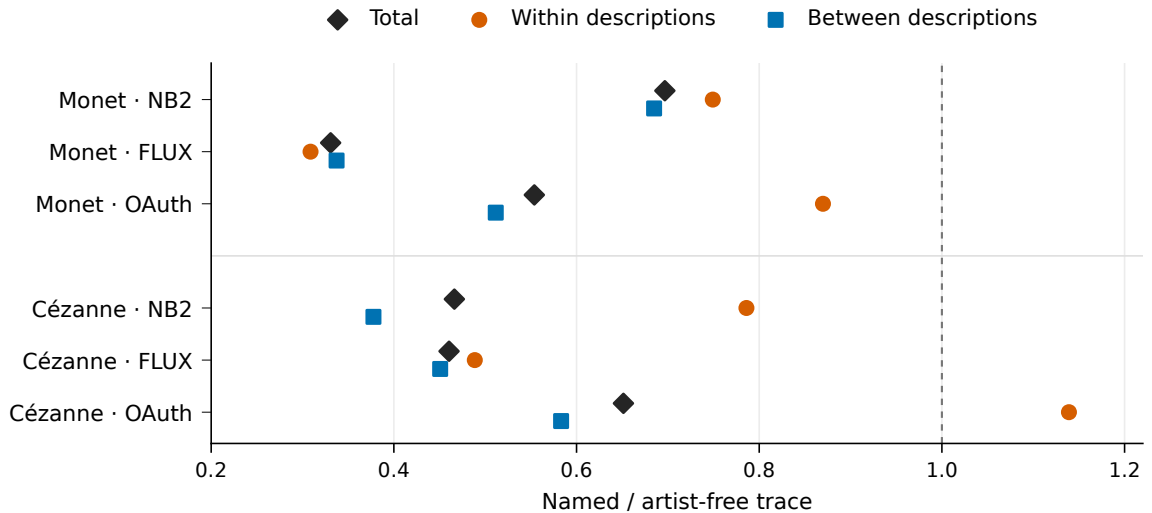
The named/artist-free total-trace ratios range from .331 to .697 in the primary representation. Decomposition reveals different sources of this contraction (Figure 2). FLUX reduces both within-description and between-description variation strongly: the respective ratios are .309 and .337 for Monet, and .489 and .451 for Cézanne. For NB2/Cézanne, within-description variation decreases to .786 while between-description variation decreases to .378. Its aggregate reduction therefore conceals a much larger relative change between descriptions than between repetitions of a description.

The OAuth/Cézanne case shows why the distinction is necessary. Total trace falls to .651 of its artist-free value even though within-description trace rises to 1.139; between-description trace falls to .583. A tighter overall cloud can therefore coexist with more variable repetitions. These observations do not identify a latent model mechanism or demonstrate fewer perceptually distinct outputs. They locate the measured variation in the actual repeated-prompt design.

## 4.3 Contraction is more consistent than proximity across feature views

All 72 named/artist-free total-trace ratios across the four feature views and three pipelines are below one, ranging from .308 to .816. Proximity is less consistent: 66 of 72 energy changes are negative. Both FLUX painter comparisons and NB2/Cézanne remain negative in all 12 view/pipeline combinations. For NB2/Monet, the 256-pixel color/spatial-only comparison instead increases energy by .416. The same no-texture view at 512 pixels gives a named/reference trace ratio of 1.101. Consequently, robust contraction relative to artist-free outputs does not imply robust below-reference spread or uniformly improved proximity.

The sensitivity has a substantive measurement basis. Texture coordinates contribute 49.0% and 47.1% of primary reference trace for Monet and Cézanne. Some multiscale coordinates are strongly correlated in development data. Agreement among such coordinates does not constitute independent evidence of the same artistic property. The diagnostic views probe this dependence without selecting a preferred representation from the observed effects.



**Figure 2:** Where generated variation changes under naming. Each point is a named/artist-free trace ratio in the primary representation. Total variation is the sum of within-description and between-description components, so its ratio need not equal their unweighted average. The line at one denotes no change. OAuth/Cézanne contracts in total while increasing within-description variation. These finite-set decompositions are descriptive.



Deleting individual reference works, holding-collection groups or content classes preserves the direction of all four NB2/FLUX primary proximity changes. These directions also survive omission of any one development painter from scaling. Some OAuth changes cross zero under these perturbations. Deletion stability supports the paid-service comparisons within the available panel, but does not correct shared reproduction differences or turn that panel into an oeuvre sample.

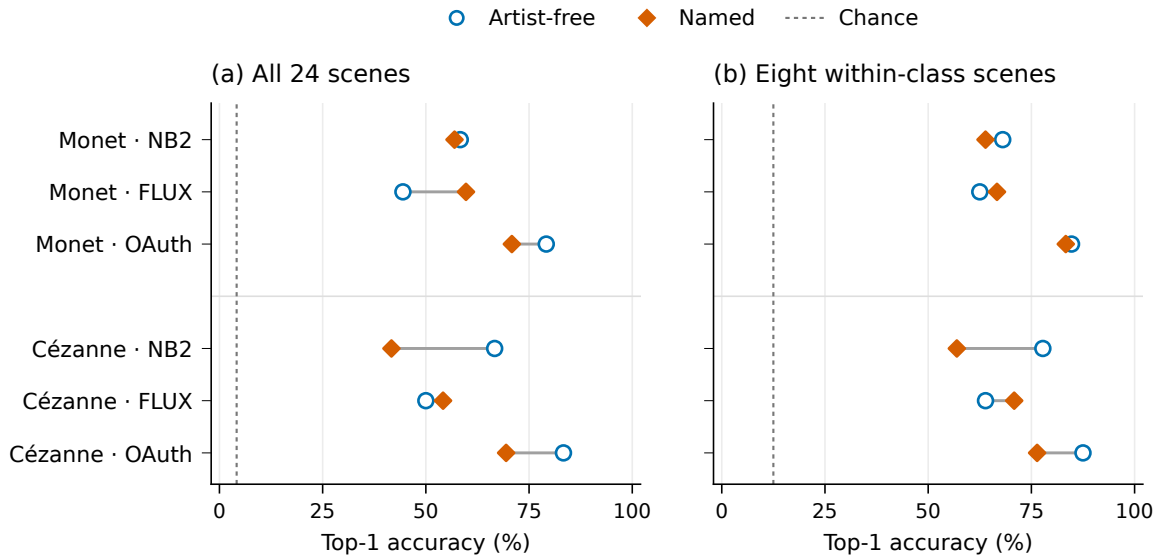
#### 4.4 Contraction need not reduce scene distinguishability

In the primary 31-feature retrieval diagnostic, naming improves accuracy in two of six service–painter cells and lowers it in four (Figure 3). FLUX/Monet improves from 44.4% to 59.7%, although its equal-scene between-description trace falls to .336 of the artist-free value. Its retrieval gain remains positive under all three pipelines, including within-class candidate restriction. This is a direct counterexample to inferring lost scene distinguishability from contraction alone.

The changes are heterogeneous. For Cézanne, accuracy declines from 66.7% to 41.7% on NB2 and from 83.3% to 69.4% on OAuth. Both declines persist across all five feature views and three pipelines in both candidate settings. For Monet, the OAuth decline is smaller within class (84.7% to 83.3%) than across all 24 scenes (79.2% to 70.8%). The NB2 difference is one correct query out of 72. Changes in separability therefore vary across services and painters.

#### 4.5 Generic painting language also reduces color response

All four Study 2 arms respond positively to vivid rather than muted instructions (Figure 4). The vivid-minus-muted differences are 3.489, 2.636, 2.351 and 2.618 development-IQR units for the free, generic, Monet and Cézanne arms, respectively; all nominal 95% response intervals are above zero. The secondary generic-minus-free contrast is  $-0.853$ , with nominal interval  $[-1.136, -0.570]$ , a 24.5% reduction relative to the artist-free point estimate. Named-minus-free



**Figure 3:** Held-repetition scene retrieval in Study 1. Each condition has 72 queries; centroids use only the other two repetitions. Panels compare 24 scene candidates with eight candidates in the same broad class. All scenes have equal query weight. Dashed lines mark chance rates of  $1/24$  and  $1/8$ , respectively. These are descriptive accuracies in the primary 31-feature representation, not semantic-adherence scores or independent scene-population estimates.

contrasts are  $-1.137$  for Monet and  $-0.871$  for Cézanne, with nominal intervals  $[-1.375, -0.900]$  and  $[-1.062, -0.679]$ . The named-minus-free contrast alone therefore cannot show that the reduction is specific to painter naming.

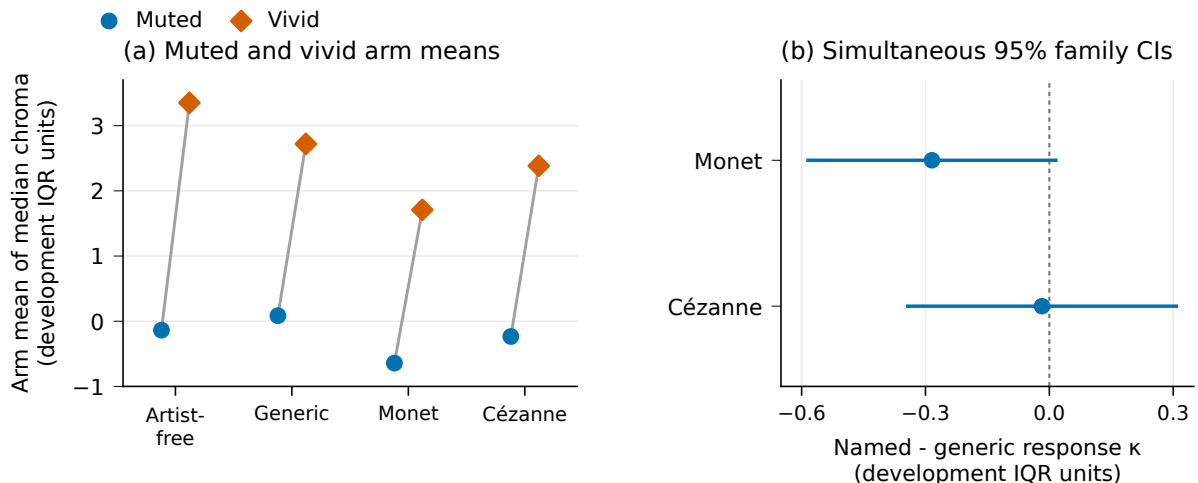
The additional painter-minus-generic interactions are  $\hat{\kappa}_M = -0.284$  (simultaneous interval  $[-0.585, 0.016]$ , Holm  $p = .0649$ ) and  $\hat{\kappa}_C = -0.018$  ( $[-0.343, 0.308]$ ,  $p = .8886$ ). Neither primary comparison resolves the direction. Monet’s six scene-specific point estimates are all negative, whereas Cézanne’s have three positive and three negative signs. Non-rejection does not establish equivalence or rule out smaller interactions.

The Monet estimate is similar at 256 pixels ( $-.301$ ) and after JPEG90 processing ( $-.308$ ). The JPEG interval narrowly excludes zero, with upper endpoint  $-.004$ , whereas the primary and 256-pixel intervals have upper endpoints  $.016$  and  $.003$ . This processing-dependent boundary crossing does not replace the prespecified primary result. Cézanne remains unresolved under both variants. The arm means also separate color level from responsiveness: relative to generic language, Cézanne lowers the muted and vivid levels by  $.319$  and  $.337$  IQR units, respectively, while leaving their average contrast nearly unchanged. Two levels cannot identify whether such patterns arise from a shifted operating range, response gain or saturation.

#### 4.6 Similar means and broad ranges can mask different chroma distributions

The named arms’ equal-polarity mixtures have means relatively close to the original reference means, yet assign their mass differently (Table 2). Most reference paintings lie inside the generated central range, while most generated outputs lie outside the reference central range. Broad range inclusion therefore does not establish a matched distribution.

Equal broad-class reference weighting gives named-mixture Wasserstein distances of  $.603$  for Monet and  $.759$  for Cézanne, compared with  $1.071$  and  $.787$  for the same generic-control mixture. Naming improves this comparison in both cases, with directions preserved across all three pipelines and both reference weightings. The results therefore do not support a general claim that naming makes the original chroma distribution harder to approach. Nor do the two extreme instructions establish continuous coverage of the intervening range.



**Figure 4:** Study 2 color levels and additional painter-name interactions. Left: equal-scene mean image-median chroma, with 24 images per arm/polarity; these points have no uncertainty bars. Right: named-minus-generic differences in the vivid-minus-muted response, with approximate simultaneous 95% family intervals (Bonferroni marginal coverage 97.5%). Both panels use the same fixed chroma IQR unit but display different estimands. The two interactions reuse the generic control observations.

| Painter | Mean chroma |           | Mass inside central range |               | $W_1$ |
|---------|-------------|-----------|---------------------------|---------------|-------|
|         | Original    | Generated | Gen. in orig.             | Orig. in gen. |       |
| Monet   | .369        | .533      | 37.5%                     | 89.5%         | .620  |
| Cézanne | .812        | 1.075     | 16.7%                     | 93.8%         | .716  |

**Table 2:** Chroma distributions in Study 2 relative to the empirical reference panel. Generated values use each painter’s 48 named outputs with equal class and polarity mass. Central ranges are empirical 10th–90th percentiles, not confidence intervals. Means and Wasserstein-1 distances use development-IQR units. The forced muted/vivid mixture is not an unprompted sampling distribution.

## 5 Discussion

Across these painter cases, adding a name changes distributional proximity and the organization of variation. Study 1 shows that a generated collection can move closer to its reference panel while contracting relative to artist-free outputs. The decomposition prevents interpreting that contraction as a uniform reduction in variation between repetitions. The retrieval diagnostic further shows that even substantial between-scene contraction can coexist with better scene distinguishability. Distances, variance and distinguishability answer related but different questions.

The controlled experiment tests whether a painter clause attenuates the color contrast relative to a generic clause. Generic painting language already produces a substantial decrease relative to artist-free prompts, and the two named-minus-generic interactions remain unresolved. Thus named-versus-artist-free attenuation does not by itself isolate a painter-specific effect. The observed shifts in color level motivate a more specific account of how style clauses alter the range of responses to palette instructions. They do not identify attention competition, a latent style prior or a training-data mechanism: those are possible explanations requiring further interventions.

Connecting these results to original paintings also requires separating response capacity from the probability mass assigned to different outputs. The forced muted/vivid mixtures can place many reference values within their broad range without approximating the reference density. Conversely, naming improves the mixture’s chroma distance relative to generic language. These findings do not establish that attenuation prevents approaching the reference chroma distributions. They do not show that intermediate palettes are unreachable, because no intermediate instruction was tested.

Several limitations bound the interpretation. The reference panel is selected, small and uneven across content classes; its digital reproduction workflows and fine-content matching are unresolved. The representation is interpretable but not a validated model of style. Texture has substantial influence, and the NB2/Monet proximity reversal under a no-texture processing variant rules out a representation-independent claim. Reused images across sensitivity analyses do not supply independent replications. Study 2 concerns one deployed service and six fixed scenes, with approximate small-repeat inference; no new-scene or cross-service interaction is estimated.

Delivery also forms part of the treatment response. In Study 1, square paid outputs contrast with nonsquare references, making high domain-detection accuracy ambiguous as evidence about style (Appendix D). The OAuth conditions in both studies differ in returned geometry and reported quality. Conditioning on these post-request fields would change the estimand and could select outputs. We therefore retain them and interpret the complete service response, without claiming a semantic-clause effect at matched rendering settings. Human judgments and independent reference validation were not collected, so the computational findings do not establish perceptual fidelity or artistic value.

A focused extension would sample intermediate palette instructions and multiple generic style phrasings under a new prospective design. This could distinguish level shifts from changes in

response slope and test dependence on the control sentence. Replication across new scenes and services would address transfer, while independently curated reproductions and perceptual evaluation would address the relation of the measured features to artistic style. These are separate uncertainties; increasing the present cells alone would not resolve all of them.

## 6 Conclusion

Painter naming often improves primary feature proximity while reducing aggregate variation in these generated collections. Contraction alone establishes neither poorer scene distinguishability nor attenuation beyond generic style language. Scene retrieval can improve under contraction, generic painting language already changes color response, and the two named-minus-generic interactions remain unresolved. Retrieval is invariant to uniform rescaling, so its improvement does not exclude a reduction in absolute response gain. Distribution-based evaluation should therefore separate proximity, variation, distinguishability and response to controlled instructions, while treating the reference panel and delivered service as part of the measurement context.

## Data and code availability

The accompanying repository contains the exact prompt inventory, reference identities and annotations, standardized feature vectors, analysis code, complete numeric tables and figure-generation code. The two study reports and descriptive diagnostics retain their separate pre-specified endpoints, acquisition accounting and sensitivity results. Weighted empirical quantiles use the exact-CDF correction documented with the data. The manuscript README gives their locations and offline reproduction commands. Raw painting reproductions and full-resolution generated images are retained separately and are not included in the repository’s redistributed files. Manuscript figures can be rebuilt from the redistributed numeric tables. Full integrity-checked analysis replay additionally requires the separately retained response archive, but makes no new API calls.

## A Prompt and feature specification

An example detailed named prompt is:

Create an oil painting on canvas. In the style of Claude Monet. Scene: A pond viewed from close to its edge, with groups of floating lily pads, gaps of open water and reflections of surrounding trees. The water fills the scene; no horizon or prominent person is visible. Render only the painting area, without a surrounding frame, signature, letters or watermark.

The artist-free version removes only the painter-style sentence. The short-scene named version substitutes “A pond with water lilies.” for the detailed scene. All 24 descriptions are available in the accompanying prompt inventory.

Study 2 uses six separate fixed descriptions, covering a river and river bend, a village and houses, and fields and a garden. The common opening and closing instructions are retained. The style sentence is absent, generic (“In a traditional landscape-painting style.”), Monet or Cézanne. The palette sentence is “Use a muted, low-chroma palette throughout the painting.” or “Use a vivid, high-chroma palette throughout the painting.” Each of the 48 scene/style/palette combinations has four outputs. The exact six scene texts and complete request inventory accompany the analysis.

| Family          | Count | Coordinates                                                                                                                                    |
|-----------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Color           | 11    | Lightness median/IQR; chroma median/IQR; chromatic fraction; hue concentration/entropy; chromatic differences at three scales and their slope. |
| Spatial         | 8     | Spectral slope/residual/anisotropy; orientation entropy; horizontal-vertical balance; gradient median/IQR; quadrant divergence.                |
| Digital texture | 12    | Four wavelet energies, their slope and curvature; three local-binary-pattern entropies; local coefficients of variation at three scales.       |

**Table 3:** The fixed interpretable representation. Coordinates describe digital image statistics, not physical paint relief. The 28-coordinate view removes chromatic-difference slope, wavelet slope and wavelet curvature.

## B Paired randomization statistic

Let  $A_i, B_i$  be a retained pair with common weight  $b_i$ , and define  $C(v) = \sum_j a_j \|v - x_j\|_2$  and  $T(v) = \sum_i b_i (\|v - A_i\|_2 + \|v - B_i\|_2)$ . Then

$$c_i = b_i \{2[C(B_i) - C(A_i)] - [T(B_i) - T(A_i)]\}, \quad \sum_i c_i = \widehat{\mathcal{E}}(X, B) - \widehat{\mathcal{E}}(X, A). \quad (5)$$

Complementary pair swaps produce  $\sum_i s_i c_i$ ,  $s_i \in \{-1, 1\}$ , exactly, including the within-generated distances. Independent randomized condition positions justify the sign assignments under the stated sharp null. Repeated scene descriptions are not treated as independently sampled scenes for population inference. With  $e$  sign draws at least as extreme in absolute value as the observed statistic, including ties,  $p = (1 + e)/100,000$ . All contrasts retain 24 distinct descriptions; the two missing short-scene outputs reduce only the Cézanne detailed/short-scene comparison to 70 pairs.

## C Color experiment: estimation and acquisition

For fixed scene  $j = 1, \dots, J$ , repetition  $r = 1, \dots, R$  and arm  $a$ , let  $D_{jra} = z_{jra,+} - z_{jra,-}$  and  $K_{jr} = (D_{jrM} - D_{jrG}, D_{jrC} - D_{jrG})^\top$ , with  $J = 6$  and  $R = 4$ . Writing  $\bar{K}_j = R^{-1} \sum_r K_{jr}$  and  $S_j$  for their within-scene sample covariance, the estimates and covariance are

$$\hat{\kappa} = \frac{1}{J} \sum_j \bar{K}_j, \quad \widehat{\text{Cov}}(\hat{\kappa}) = \sum_j \frac{S_j}{J^2 R}. \quad (6)$$

This retains the off-diagonal covariance from shared generic controls. For endpoint  $p$ , let  $v_{jp} = S_{j,pp}/(J^2 R)$ . The approximate degrees of freedom are

$$\nu_p = \frac{(\sum_j v_{jp})^2}{\sum_j v_{jp}^2 / (R - 1)}. \quad (7)$$

The two simultaneous intervals are  $\hat{\kappa}_p \pm t_{1-.05/(2.2), \nu_p} \sqrt{\sum_j v_{jp}}$ . Two-sided  $t$   $p$ -values receive Holm adjustment within this two-endpoint family. The scenes are fixed; between-scene variation of mean responses is not added as a random-scene variance component. Estimated scene means retain generation noise. The assumptions do not follow from dispatch randomization alone.

An earlier allocation was stopped after 49 images and one complete HTTP 503 response, leaving 142 assigned slots unstarted. A narrowly specified transport correction and a replacement collection using the same fixed prompts and order were designated before successful images were visually inspected or their scientific features measured. The 192-image replacement is the sole Study 2 inferential cohort. The 49 earlier images were measured after replacement collection

ended and remain ancillary, without pooling or filling missing slots. Their incomplete assigned grid receives no primary inference. All 192 replacement requests succeeded, with no failures, retries or missing measurements.

## D Supporting Study 1 distribution diagnostics

### D.1 Relative painter alignment

Painter specificity uses equal one-third content-class weights for both painters. With  $G_M, G_C$  denoting collections assigned to Monet and Cézanne, define

$$I = \hat{\mathcal{E}}(X_M, G_M) + \hat{\mathcal{E}}(X_C, G_C) - \hat{\mathcal{E}}(X_M, G_C) - \hat{\mathcal{E}}(X_C, G_M). \quad (8)$$

Negative  $I$  favors own-painter over cross-painter alignment. The within-reference and within-generated terms cancel, making this a contrast of relative alignment. Artist-free prompts are identical across the two collection labels and provide a descriptive control for their arbitrary painter assignment.

Under equal content-class weights, naming also strengthens relative own-painter alignment. The interaction  $I$  changes from .101 to  $-.855$  for NB2, from  $-.048$  to  $-1.755$  for FLUX and from  $-.044$  to  $-1.802$  for OAuth. Its named-minus-free change is negative in all 36 service/view/pipeline combinations. This joint contrast does not require each generated collection individually to be closest to its own painter’s references. Separately generated artist-free collections have identical prompt payloads across painter labels yet nonzero empirical discrepancies. These controls illustrate finite-set and collection variability; they do not provide an equivalence threshold.

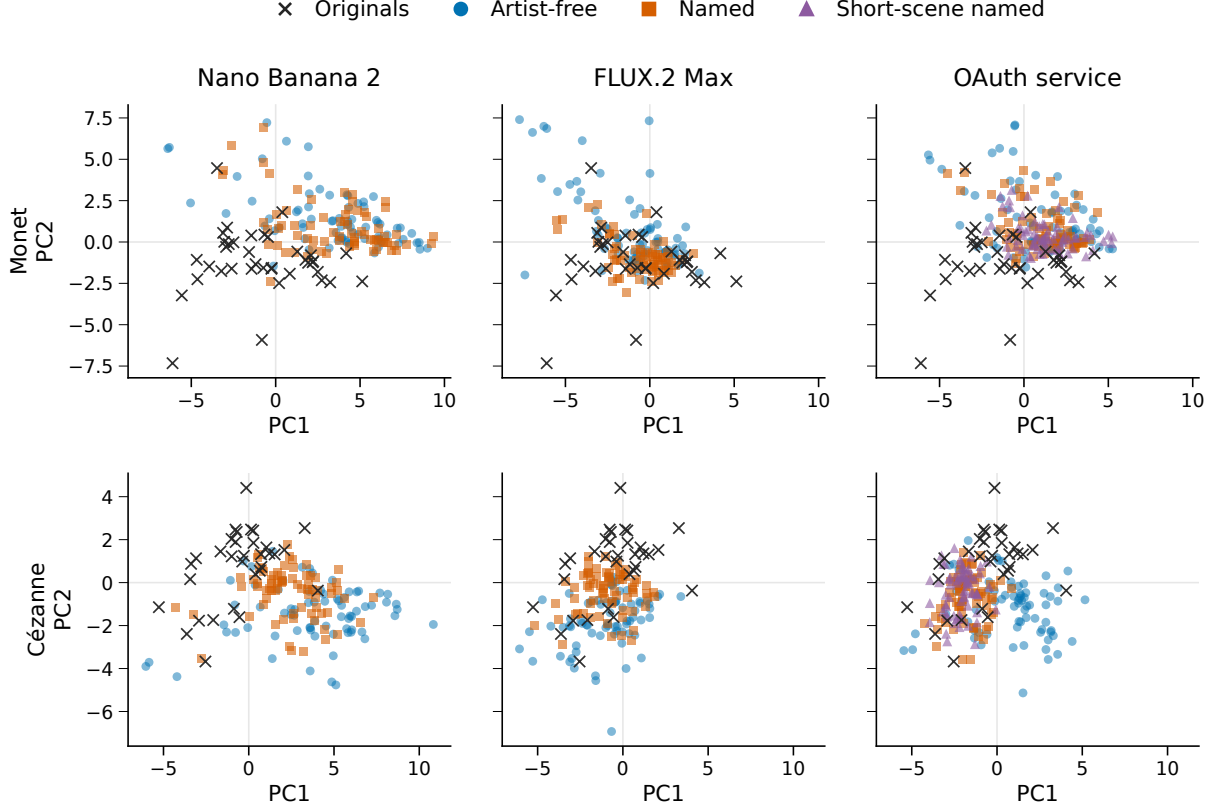
### D.2 Projection and domain detection

For visualization, we fit one common principal-component analysis (PCA) per painter, assigning half the weight to references and sharing half equally across the seven generated cells, with content weighting within each. All cells use that painter’s common axes. Full-space linear and radial-basis-function (RBF) kernel-ridge classifiers use six folds holding out entire generated descriptions and disjoint original works, without tuning or PCA inputs. This grouping respects the repeated-prompt structure (Roberts et al., 2017). Cross-service evaluation trains on one service and tests on another using the same held-out description/work scheme. We report balanced accuracy at a fixed zero decision threshold and mean within-fold AUC separately; the former also depends on calibration.

The common PCA views show overlapping clouds with changes in location and concentration (Figure 5). The first two components retain 48.4% of weighted variation for Monet and 46.0% for Cézanne, leaving more than half outside the plots. Visual overlap consequently cannot establish equality of the full distributions. Conversely, separation in a projection would not establish a difference in artistic style.

In the full feature space, within-service RBF balanced accuracy ranges from .893 to .987 across the 14 original/generated comparisons. However, all 576 NB2/FLUX images are square and all 70 references are nonsquare: a rule using only square geometry separates these domains perfectly. Capture workflows for the reference reproductions are also unresolved. High classification accuracy therefore establishes differences between the recorded image domains, not a validated stylistic distinction.

Cross-service behavior further limits detector interpretation. At the fixed threshold, naming lowers balanced accuracy in every directed service/painter comparison for both kernels. Named-condition ranges are .410–.874 for the linear kernel and .447–.704 for RBF. This need not mean



**Figure 5:** Original and generated feature distributions in common painter-specific PCA coordinates. Every Study 1 image is shown; references are repeated across service panels for comparison. Axes and limits are shared within each painter, but the two painters have separate PCA fits. The short-scene named condition is available only for OAuth. Projections show overlap and concentration while retaining less than half of total weighted variation.

uniformly weaker ranking discrimination. For Monet with training on FLUX and testing on OAuth, linear balanced accuracy falls from .904 to .874 while mean within-fold AUC rises from .953 to .968. The observed threshold transfer includes calibration changes; below-chance scores are not evidence that generated images have become more authentic.

### D.3 Classifier specification

For  $d = 31$  standardized coordinates, the fixed kernels are  $K_L(u, v) = u^\top v / d + 1$  and  $K_R(u, v) = \exp(-\|u - v\|_2^2 / d) + 1$ . The classifiers minimize weighted squared error with a kernel penalty of .01, using equal total training weight for original and generated labels and the reference content proportions within each. Training and evaluation memberships are fixed. Sparse reference classes mean not every test fold contains all three classes; pooled balanced accuracy uses the complete held-out set. For cross-service evaluation, AUC is computed within each fold and then averaged across six folds, rather than pooling margins from differently trained classifiers. Complete kernels, memberships and scores are included with the analysis code and numeric tables.

The accompanying diagnostic tables retain the complete cross-service and reference-coverage sensitivity analyses.



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